

Department of Electrical and Computer Engineering  
University of Wisconsin – Madison  
ECE 552 Introduction to Computer Architecture  
**In-Class Exercise (10/03)**

1. You are given the following sequence of MIPS instructions.

```
L1:    or  $s1, $s2, $s3
L2:    or  $s4, $s1, $s6
L3:    or  $s1, $s2, $s3
L4:    or  $s3, $s4, $s5
```

- a) Identify all **register dependences** in this sequence. For each dependence, state the **register, instruction pair** and **dependence type**: read-after-write (RAW), write-after-read (WAR) and write-after-write (WAW). One dependence has been provided below as an example.

RAW (\$s1): L1 → L2

**Answer:**

**WAW (\$s1): L1 → L3**

**WAR (\$s3): L1 → L4**

**WAR (\$s1): L2 → L3**

**RAW (\$s4): L2 → L4**

**WAR (\$s3): L3 → L4**

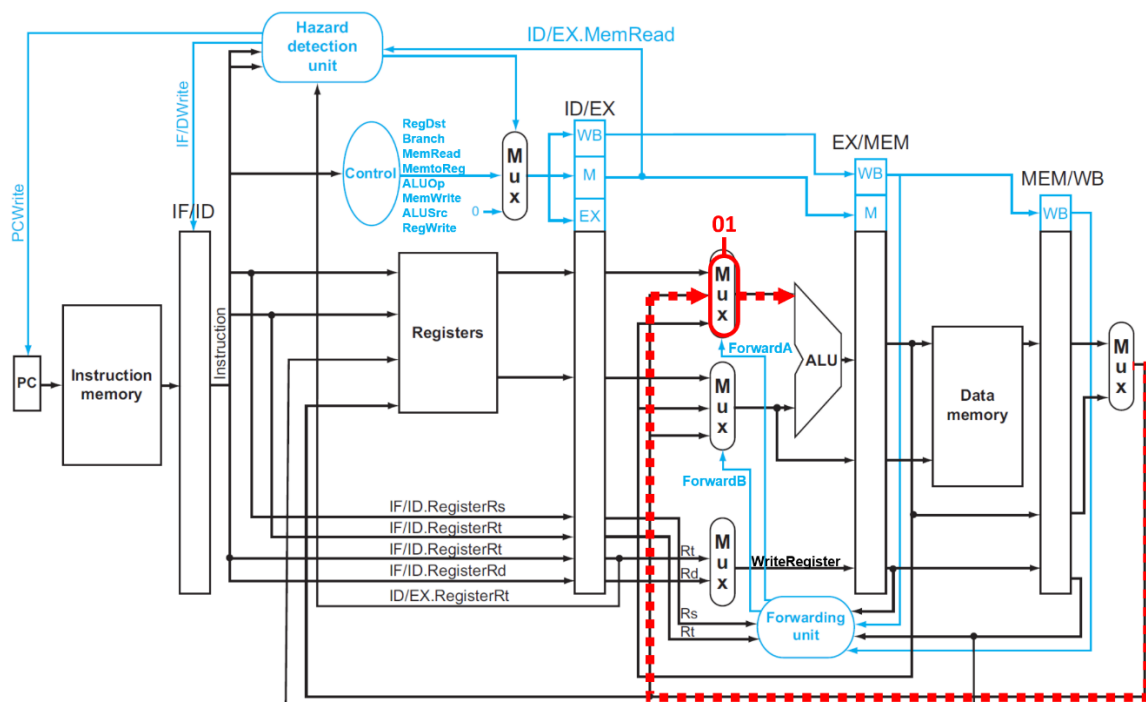
- b) Assume the above sequence of instructions is executed on our five-stage pipeline **without** support for data forwarding nor RF bypassing. What is the minimum number of **stall cycles (i.e., nops)** needed for this program to run correctly? Fill in the **pipeline diagram** below to show how you got your answer. For example, if W is stalling on cycle 8 but executing on cycle 9, then cycle 8 would contain **W\*** and cycle 9 would contain **W**. Then sum the number of cycles that contain stalls.

| instruction | clock cycle |   |    |    |    |   |   |    |    |    |    |    |    |    |
|-------------|-------------|---|----|----|----|---|---|----|----|----|----|----|----|----|
|             | 1           | 2 | 3  | 4  | 5  | 6 | 7 | 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| L1          | F           | D | X  | M  | W  |   |   |    |    |    |    |    |    |    |
| L2          |             | F | D* | D* | D* | D | X | M  | W  |    |    |    |    |    |
| L3          |             |   | F* | F* | F* | F | D | X  | M  | W  |    |    |    |    |
| L4          |             |   |    |    |    |   | F | D* | D* | D  | X  | M  | W  |    |
|             |             |   |    |    |    |   |   |    |    |    |    |    |    |    |
|             |             |   |    |    |    |   |   |    |    |    |    |    |    |    |
|             |             |   |    |    |    |   |   |    |    |    |    |    |    |    |
|             |             |   |    |    |    |   |   |    |    |    |    |    |    |    |

**Answer:**

**5 cycles = 3 for RAW (\$s1) + 2 for RAW (\$s4)**

2. Consider our five-stage pipelined processor with **full forwarding** and **RF bypassing**. The figure below shows a **specific MEM-to-EX forwarding path** in our pipeline (in red dashed lines, based on COD Figure 4.60).



- a) Fill in the blanks below, specifying the **condition logic** used by the **Forwarding Unit** to determine when to enable this specific MEM-to-EX forwarding path.

```

if ( ... ) // EX-to-EX forwarding conditions, omitted here
    ForwardA = 10;

else if ( (MEM/WB._____ == 1)
    && (MEM/WB.WriteRegister != 0)
    && (MEM/WB.WriteRegister == _____) )
    ForwardA = 01;

else
    ForwardA = 00;

```

**Answer:**

```

else if ( (MEM/WB.RegWrite == 1)
    && (MEM/WB.WriteRegister != 0)
    && (MEM/WB.WriteRegister == ID/EX.RegisterRs) )
    ForwardA = 01;

```

b) Assume the following sequence of instructions is executed on this processor:

```

L1:  lw    $s1, 0($s2)
L2:  or     $s3, $s1, $s4
L3:  andi   $s3, $s3, 255
L4:  sub    $s1, $s1, $s4
L5:  add    $s5, $s3, $s6

```

While executing these instructions, the specific MEM-to-EX forwarding path above is enabled **at least once**. If instruction L1 is in the fetch (IF) stage at cycle 1, during **which cycle(s)** is this MEM-to-EX forwarding path enabled (i.e., when is the red-highlighted mux's select signal ForwardA equal to 01)? Fill in the **pipeline diagram** below to show how you got your answer.

| instruction | clock cycle |   |    |   |   |   |   |   |   |    |    |    |
|-------------|-------------|---|----|---|---|---|---|---|---|----|----|----|
|             | 1           | 2 | 3  | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| L1          | F           | D | X  | M | W |   |   |   |   |    |    |    |
| L2          |             | F | D* | D | X | M | W |   |   |    |    |    |
| L3          |             |   | F* | F | D | X | M | W |   |    |    |    |
| L4          |             |   |    |   | F | D | X | M | W |    |    |    |
| L5          |             |   |    |   |   | F | D | X | M | W  |    |    |
|             |             |   |    |   |   |   |   |   |   |    |    |    |
|             |             |   |    |   |   |   |   |   |   |    |    |    |
|             |             |   |    |   |   |   |   |   |   |    |    |    |

**Answer:**

**Cycles 5 and 8**